SAKTHI VEL C

Python Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science student and Python Full-Stack Developer specializing in Machine Learning and IoT, focusing on architecting secure, scalable systems. Proven expertise in building real-time AI applications with Gemini 1.5 Pro, FastAPI, and Next.js, while optimizing database performance and implementing robust encrypted data architectures for high-performance environments.

EDUCATION

Bachelor of Engineering in Computer Science — **2023 – 2027**

Kamaraj College of Engineering and Technology, Virudhunagar, India | GPA: 7.44 (till 5th sem)

TECHNICAL SKILLS

Languages & Databases: Python, C, PostgreSQL, Supabase, SQL

Development Tools & Frameworks: Next.js 14, Flutter, Tailwind CSS, FastAPI, Flask, RESTful APIs, JWT, Supabase Auth

AI & Machine Learning: Gemini 1.5 Pro, Random Forest, XGBoost, CNN, TensorFlow

IoT & Hardware: ESP Modules, Arduino IDE, MQTT, Sensor Integration

Security & Tools: Git, GitHub, RBAC, Jupyter Notebook

INTERNSHIP EXPERIENCE

Software Developer Intern — **Dec 2025 – Present**

PENTAGON GARMENTS, Virudhunagar, India

- Developed a secure verification system with RESTful APIs and JWT authentication for real-time security monitoring.
- Independently built and deployed the Security Rounds Verifier system into production for the company, currently in active daily use by their security staff.
- Received a certification from Pentagon Garments for successful completion and real-world deployment of the project.

Machine Learning Intern — **June 2025 – Nov 2025**

PANITH INNOVATIONS, Remote

- Developed and deployed predictive Machine Learning models using XGBoost and Random Forest to enhance system accuracy and project performance.

KEY PROJECTS

KURAL – AI Voice-Based Grievance Redressal System | Gemini 1.5 Pro, PSTN — **Jan 2026 – Present**

- Engineering a PSTN-based platform using Gemini 1.5 Pro to convert offline voice calls into structured digital reports.
- Implementing speech recognition and automated intent detection to classify and escalate grievances to authorities.

Security Rounds Verifier | FastAPI, Next.js, PostgreSQL, JWT — **July 2026**

- Architected a full-stack verification system with Role-Based Access Control (RBAC) and real-time monitoring.
- Optimized database queries reducing response time by 40% for high-concurrency environments.
- Deployed and hosted the system in production for Pentagon Garments; actively used by the company for real-world security monitoring. Live at apps.pentagontextiles.com | Code: github.com/sakthivel-136/veripatrol.
- Awarded a certification by Pentagon Garments for successful project completion and live deployment.

IoT Smart Air Quality & Climate Monitoring System | ESP32, Random Forest — **Nov 2025**

- Developed a predictive alert system using Random Forest to detect health risks and HVAC inefficiencies.
- Presented research findings at the ICASET-2026 International Conference in Chennai.

CONFERENCES & ACHIEVEMENTS

- **Presenter** – 5th International Conference on Advances in Science, Engineering & Technology (ICASET-2026). Presented research on IoT-integrated Air Quality Systems.
- **Patent Filed** – Intellectual Property application filed for ”Mr. Strict” (AI Proctoring System).
- **Grand Winner** – Student Pitch Competition at Mini Technovision 2K26.
- **ISRO Certification** – Completed AIML training for Crop Acreage Mapping via official ISRO workshop.
- **Industry Certification** – Certified by Pentagon Garments for developing and deploying the Security Rounds Verifier system into real-world production use.